

Retail Sales Analysis using Exploratory Data Analysis (EDA)

1. Objective

The main objective of this project was to perform Exploratory Data Analysis (EDA) on the Retail Sales dataset. The analysis was carried out to understand the dataset, identify data quality issues, study relationships between different variables, and gain meaningful insights about customer purchasing behavior and sales patterns.

2. Dataset Description

The Retail Sales dataset contains transaction records from a retail store. Each record represents a customer's purchase and includes details such as customer information, product category, quantity purchased, price per unit, and total amount spent.

Dataset Features:

- Transaction ID
 - Date
 - Customer ID
 - Gender
 - Age
 - Product Category
 - Quantity
 - Price per Unit
 - Total Amount
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3. Data Exploration

The dataset was explored using the following functions:

- `head()` to view the first five records.
- `info()` to check data types and missing values.
- `shape` to know the number of rows and columns.
- `columns` to view all feature names.
- `describe()` to obtain summary statistics of numerical columns.

This helped in understanding the overall structure of the dataset before analysis.

4. Data Cleaning

The dataset was checked for missing values and duplicate records.

Missing Values

No missing values were found in any column.

Duplicate Records

No duplicate records were present in the dataset.

Since the dataset was already clean, no additional preprocessing was required.

5. Descriptive Statistics

Descriptive statistical measures were calculated for the numerical columns.

The following statistics were computed:

- Mean
- Median
- Mode
- Variance
- Standard Deviation

These statistics provided information about the central tendency and spread of the numerical data.

6. Correlation Analysis

A correlation matrix was calculated to understand the relationship between numerical variables.

Important Observations

- **Price per Unit** showed the strongest positive correlation (**0.851925**) with **Total Amount**.
- **Quantity** showed a moderate positive correlation (**0.373707**) with **Total Amount**.

- **Age** had a very weak negative correlation (**-0.060568**) with **Total Amount**, indicating almost no relationship.
 - **Transaction ID** had no meaningful relationship with Total Amount because it is only used as a unique identifier.
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7. Outlier Detection

The Interquartile Range (IQR) method was used to detect outliers in the **Total Amount** column.

Results:

- Lower Limit = **-1200.0**
- Upper Limit = **2160.0**

After checking the data, no values were found outside these limits. Therefore, no outlier treatment was required.

8. Filtering and Sorting

Filtering was performed to identify transactions with higher total sales values.

Sorting was applied to arrange the transactions in descending order based on **Total Amount**, making it easier to identify high-value purchases.

9. Visualizations

The following visualizations were created using Matplotlib:

- **Bar Chart:** Average Total Amount by Gender
- **Line Chart:** Average Total Amount by Product Category
- **Histogram:** Distribution of Total Amount
- **Scatter Plot:** Age vs Total Amount

These visualizations helped in understanding customer purchasing patterns and sales distribution.

10. Observations

- The dataset was clean and required no preprocessing for missing or duplicate values.
- Price per Unit had the strongest influence on Total Amount.
- Quantity also contributed positively to Total Amount.
- Age showed almost no relationship with customer spending.
- No outliers were detected in the Total Amount column using the IQR method.
- The visualizations provided a clear understanding of sales trends and customer purchasing behavior.

11. Conclusion

This project provided practical experience in performing Exploratory Data Analysis on a retail sales dataset. I learned how to explore a dataset, calculate descriptive statistics, analyze correlations, detect outliers using the IQR method, perform filtering and sorting operations, and create meaningful visualizations. The analysis showed that **Price per Unit** is the most significant factor affecting the **Total Amount**, while **Age** has very little impact on customer spending. Overall, this project improved my understanding of data analysis techniques using Python and strengthened my EDA skills.